

Resiliency Best Practices: Patterns, Practices and Pitfalls

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Definition of Resilience

The ability of an information system to continue to:

- 1) Operate under adverse conditions or stress, even if in a degraded or debilitated state, while maintaining essential operational capabilities
- 2) Recover to an effective operational posture in a time frame consistent with mission needs. **(NIST SP 800-39)**

WSJ (How to create IT Resilience article)

Simply defined, IT resilience is an organization's ability to maintain acceptable service levels, no matter what challenges arise. Because so many things can cause service disruptions —hardware failures, natural disasters, malicious attacks, and operator error, among others—creating effective disaster recovery (DR) and business continuity (BC) plans to address vulnerabilities can be daunting

Business resilience is a company's ability to overcome an unexpected business disruption and recover to **acceptable continuing operations**.

Business continuity is a set of processes and procedures that enable companies to keep running their business during a crisis.

Understanding what ‘Resilience’ mean

Case 1) ‘**I want to improve my IBM Z Resilience, from where I can start ?**’

Case 2) ‘**I had an outage**, I want to improve my IBM Z Resilience, from where I can start ?’

Our goal is to do **Case 1** ,not second!.

The resilience of your environment depends on how effectively you leverage available solutions & capabilities.

Business Continuity Patterns

Business Continuity Patterns	Pattern 1 Reliable base	Pattern 2 Failover capable	Pattern 3 Fault tolerant	Pattern 4 Adaptive continuity
Characteristics	Single-site operation Service is restored through local restart	Site-aware recovery Service is restored by restart on an additional system or at a recovery site using replicated data and redirection	Near-continuous execution Common component failures are masked, and workloads continue with remaining capacity	Policy-driven, multi-site orchestration and routing in a Failover-and-stay enabled environment
Description	The primary goal of Pattern 1 is predictable recovery . Service interruption is assumed to be acceptable, if recovery is reliable, repeatable, and operationally manageable	The primary goal of Pattern 2 is site survivability , not continuous availability. The business objective is to restore service following a system or site outage , including full site loss, by taking over processing at a designated recovery location	The primary goal of Pattern 3 is continuous service for common failures. Unlike Patterns 1 and 2, where interruption is an accepted part of recovery, Pattern 3 explicitly assumes that components will fail and designs the system to tolerate those failures without user-visible outage	The primary goal of Pattern 4 is seamless workload routing across multiple active sites or regions . Workload remain continuously available, with unplanned disruption resulting only in managed service degradation rather than service interruption or data loss
Application <i>Business logic</i>	<i>Restart-tolerant</i>	<i>Portable execution context and site-restartable design</i>	<i>Failure-transparent execution with no single-member affinity</i>	<i>Coordinated workload aware recovery</i>
Middleware <i>Database and Application Platform</i>	<i>Local restart semantics</i>	<i>Restart using replicated data sets and logs</i>	<i>Shared-state or shared-data continuity with transparent failover semantics</i>	<i>Coordinated workload aware recovery</i>
Management <i>Monitoring and automation</i>	<i>Restart automation</i>	<i>Detection, orchestration, replication control, and client redirection</i>	<i>Automation for failure detection, isolation, drain, rebalance, and reintegration</i>	<i>Policy-driven adaptive orchestration</i>
Operating system <i>z/OS, Linux on Z, etc.</i>	<i>Restart-based recovery</i>	<i>Alternate-system and alternate-site restart support</i>	<i>Coordinated concurrent execution and multisystem services</i>	<i>Coordinated execution across processing environments</i>
Infrastructure <i>Complete platforms, storage and network</i>	<i>Single-site resilient infrastructure</i>	<i>Multi-system or multi-site topology with sufficient recovery-site capacity</i>	<i>Fault-isolated redundancy across systems, coupling facilities, storage, and networks</i>	<i>Active infrastructure across distributed sites or environments</i>

Patterns are explained in detail in new Redbook.

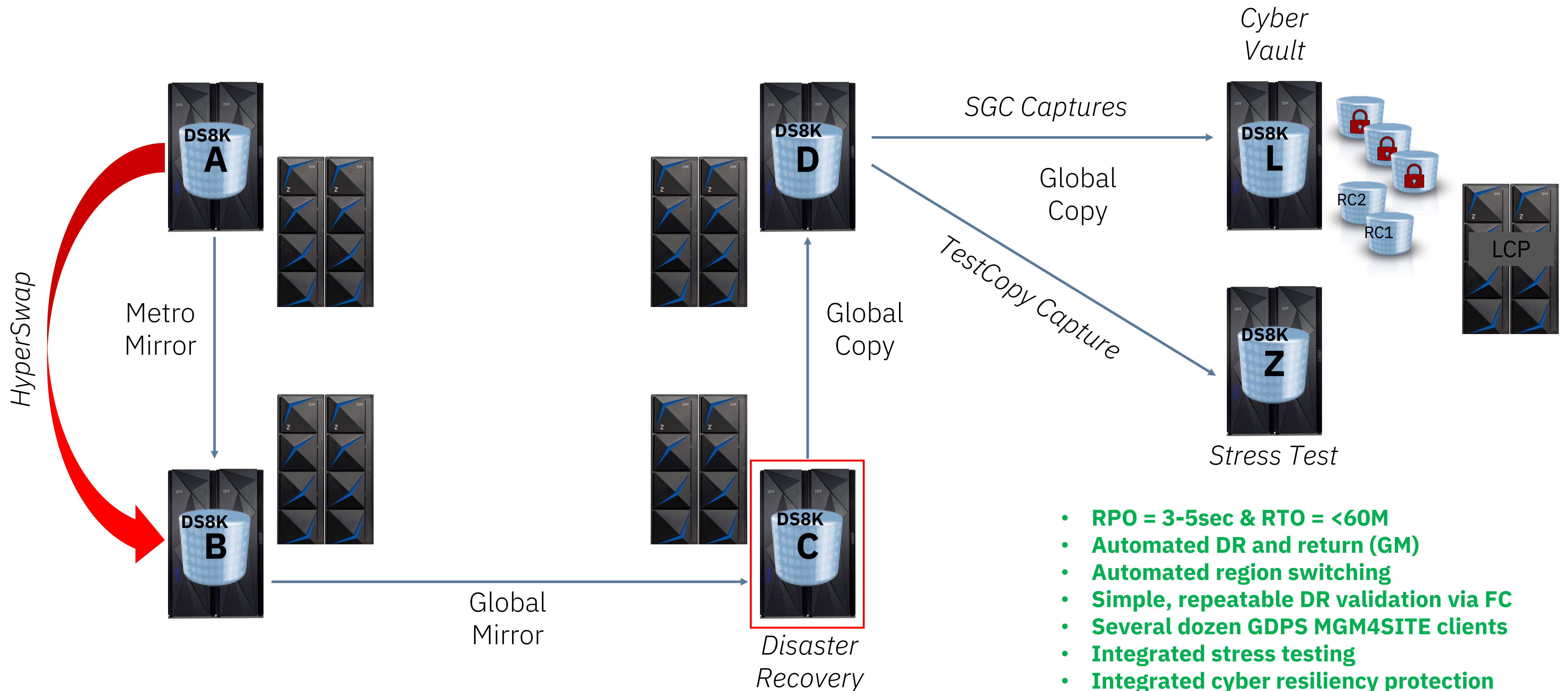
Getting Started with Business Continuity on IBM Z Redbook just published : [Getting Started with Business Continuity on IBM Z](#)

State of the Art IBM Z Configuration

State of the Art IBM Z Configuration – GDPS MGM4SITE & LCP Mgr – CA / DR / Cyber Resiliency

Active-Region

Standby-Region

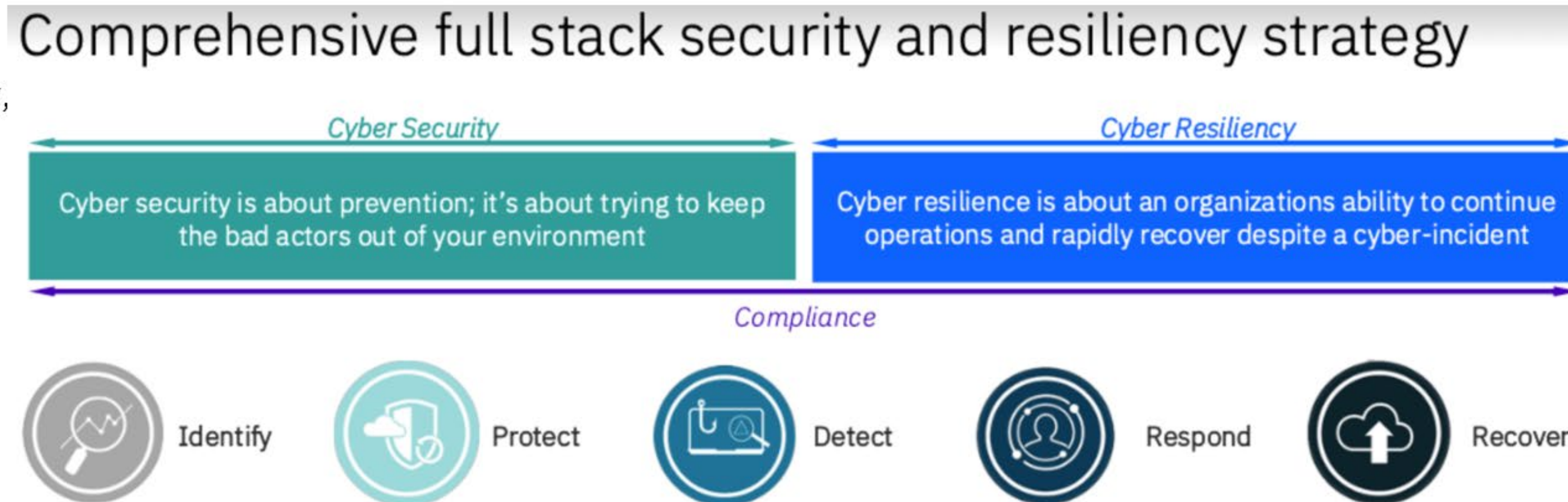


How to achieve Continues Availability ?

- ❑ Redundancy
 - ❑ Automate
 - ❑ Reduce the number of elements in critical path
-

The 5 pillars of resilience are:

- ✓ Prepare/identify,
- ✓ Protect,
- ✓ Detect,
- ✓ Respond
- ✓ Recover



Highlights of IBM Z Stack Innovations in Resiliency

Faster recovery when failures occur

- System Recovery Boost(All functions,including MiddleWare stop/restart support, Dump support)
- z/OS Automatic Restart Manager(ARM)
- Sysplex Failure Management,CFLRMGMT XCF Function
- System Status Detection Partitioning Protocol
- External CF, CF System/User Managed Duplexing
- zHyperLink (VSAM Verify) , DFHSM Dataset (z/OS & zFS File) Backup
- RLS Catalog , PDSESHARING(EXTENDED)
- z/OS AutoIPL
- IBM Z Backup Resiliency, DB2 advance recovery,IMS FDBR
- Flexible Capacity,CBU
- DVIPA,Sysplex Dynamic Workload Routing (Sysplex Distributor...)
- CICSplex,DB2,IMS Datasharing,MQ Queue Sharing
- z/OS IOS CU Recovery
- Z16 CF Improved IFCC Handling
- IBM Z Cyber Vault

Failure Avoidance

- z/OS Blocked Workload Support
- GRS Contention Analysis
- XCF Message Isolation, CFMONOPAVOID
- Message Flood Automation
- SMF Flood Automation
- Capacity Provisioning
- XCFCNFXX- LOGR Services Monitor
- Auto-WTOR Response
- IBM TST, IBM ZBURST
- Cyber Resiliency,
- IBM Thread detection for z/OS(TDz)
- IBM Concert For Z
- TFP-HW
- HMC LPAR Error Protection
- Z17 HMC Dual Control

Problem Determination and Data Capture

- Run Time Diagnostics(RTD)
- Dump cmd Parmlib members (IEADMCxx)
- AutoIPL w/Standalone Dump
- z/OSMF Incident Log
- SMF Logstreams (Performance & Scalability)
- zWorkload Information Correlator (zWIC)
- zWorkload Information Navigator (zWIN)
- New SMF Records & Fields

React faster to workload fluctuations

- zHyperLink,zHyperwrite,DFSMS Striping
- FICON 32S
- Capacity Provisioning Manager
- TFP-HW, OOCOD
- Hiperdispatch
- More Memory (Larger Buffer Pools, GBPs...)

Improved workload scaling

- Hiperdispatch
- Shared zFS
- Z17 Performance improvements
- zHyperLink, zHyperwrite,Read from Secondary, zEDC
- JVMS pause-less garbage collection
- FICON 32S,z17 CL6 Throughput improvements
- Large Page Support, RLS Catalog
- CICSplex,DB2,IMS Datasharing, MQ Queue Sharing
- XTCSIZE, zOS V1R10 VSM algorithm improvements,V1R8 RSM
- Sysplex Distributor, VSAM RLS 2 Lock Structures
- z16 CF Adapter Performance Improvements
- z16 CFCC Code Scalability Enhancements
- DFSMS Extended Addressability, DFSMS Striping

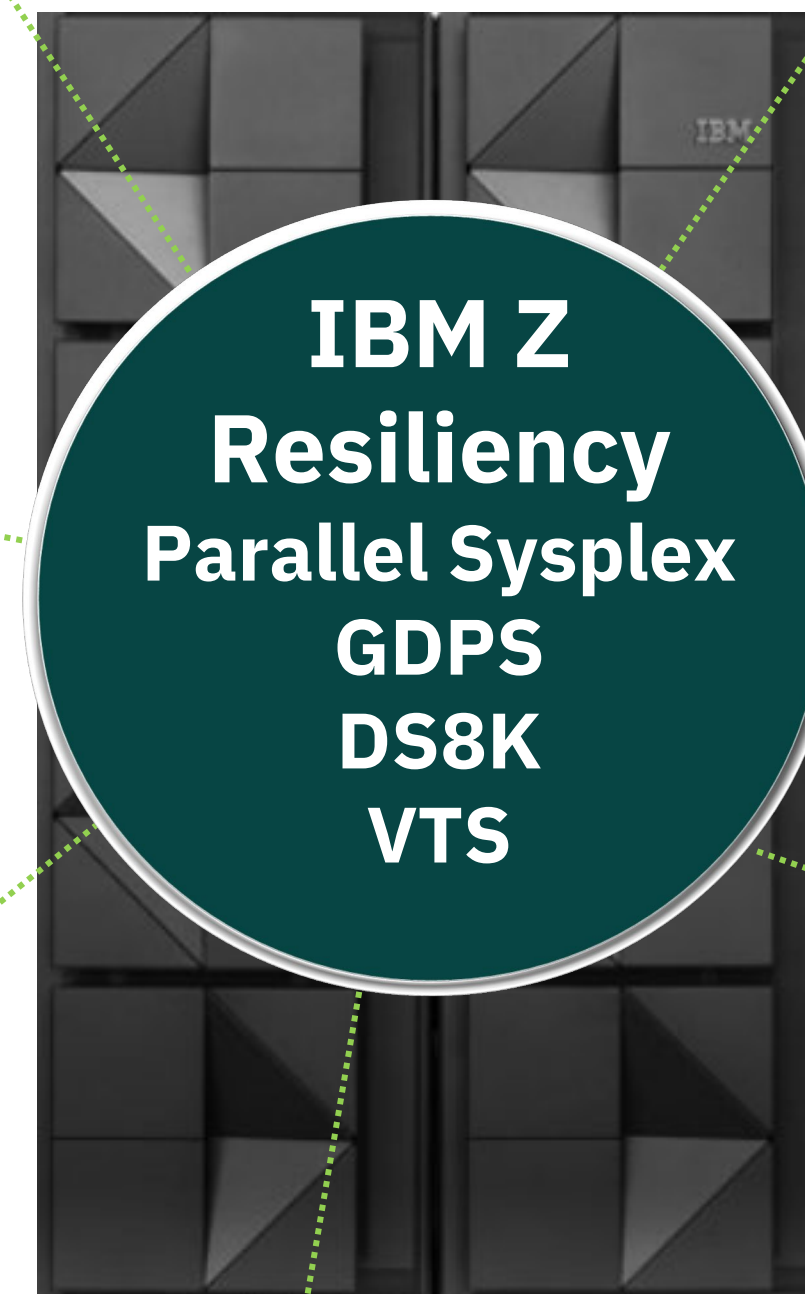
Concurrent maintenance

- Software coexistence and rolling IPLs
- z/OS concurrent maintenance, Fixcats
- Dynamic I/O Reconfiguration
- HyperSwap

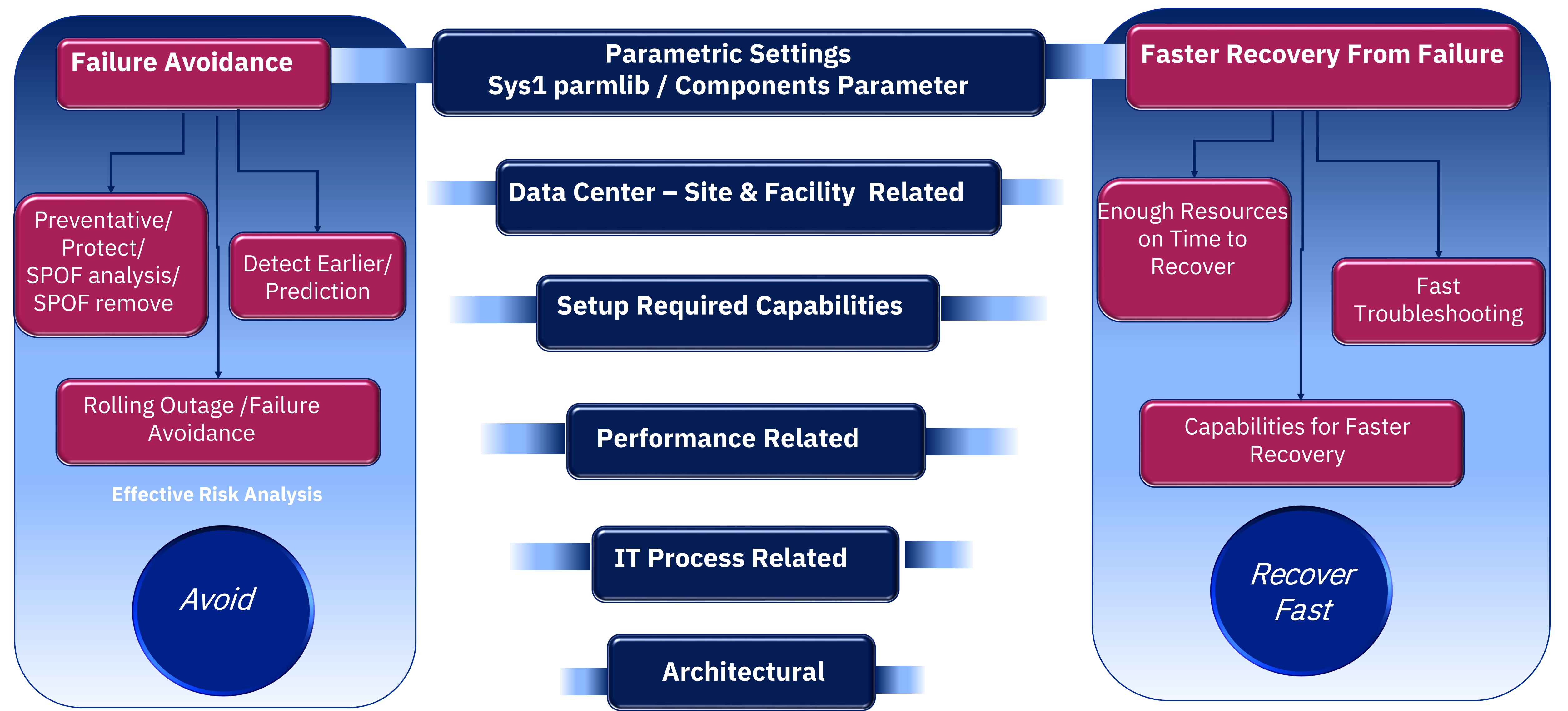
Preventative / Real-Time Insight

- z/OS Health Checker
- Predictive Failure Analysis (PFA)
- IBM Z Anomaly Analytics (IZAA now part of IBM Concert for Z)
- IBM Concert For Z
- IBM Z Operational Log and Data Analytics
- Heartbeat Monitoring – Omegamon, RMFMON, Intellimagic, OpenTelemetry

And many more...



Achieving Resiliency : System Programmer Approach



RAS= Reliability + Availability + Serviceability

Common gaps seen based on WSC Resiliency Workshops & Other Client Engagements

Data Center Site & Facility Related Outages

Architecture with SPOF (Single Point Of Failure)

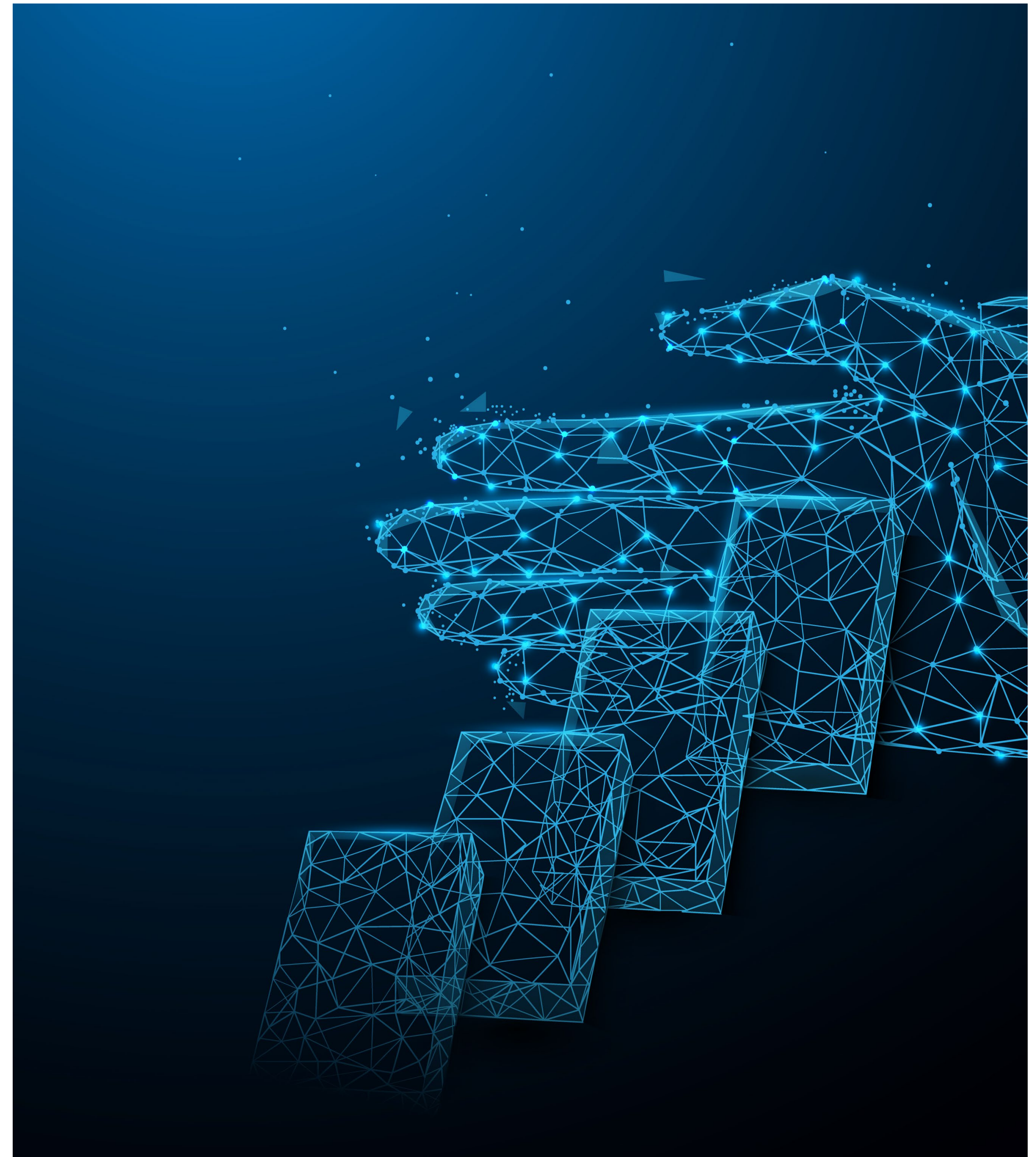
Sysplex capabilities are not enabled for all applications

IBM Z Resiliency capabilities not implemented

Correct and efficient tools to help improve the platform

Architecture not align with Business SLAs

Operational Processes Related



Resiliency Best Practices

❑ Infrastructure Architecture

- ✓ Implement Symetric Sites/Regions/ LPAR Design
- ✓ Improve DR site. Consider it as you need it anytime (not only natural distaster or DC related issue). Remember the definition of Resiliency
- ✓ Conduct DR Test Periodically, Monitor RTO and RPO values. Consider using Region Swaps, Recover all LPARs. Conduct projects to improve RTO.
- ✓ It is Not a Question of “IF”, but “WHEN” ... practice, practice, practice ... gain confidence in your CA / DR / cyber resiliency solution
- ✓ Perform DR testing - On demand (using additional copy of data) and/or Site / Region switching testing
- ✓ Prepare for cyber events – take several PiT copies of data, verify the PiT copies, and, in the event of a cyber event, restore data to production
- ✓ Leverage new capacity solutions(Ex: Flexible Capacity)
- ✓ Take advantage of IBM Advocate
- ✓ Spare LPARs in each CEC. Symmetric Design.
- ✓ Isolate non-prod LPARs from production LPARs (Different LPARs, sysplex, disk boxes...) :
Allow for testing of s/w maintenance, h/w microcode, and develop, practice, and document procedures,
Provides “hands-on” education environment.

❑ DC best practices

- ✓ Different rooms, tier level, Operational maturity level, HW power connections,....
- ✓ Resilient DR Datacenter – GDPS 4-site, all LPARs recovered
- ✓ Certified Datacenter

❑ HW management

- ✓ No SPOF related to any HW box,
- ✓ Up-to-date HW models,
- ✓ Implement Resiliency related HW features
- ✓ Maintenance strategy (Periodic maintenance), Review HIPER MCLs biweekly,
Subscribe to My Notification([My Notifications Subscription Service](#)), Leverage IBM ResourceLink
- ✓ Follow and apply maintenance included HW related Fixcats

Resiliency Best Practices

❑ LPAR level (z/OS and CF Level)

- ✓ Sysplex enabled , ALL applications support sysplex, all sysplex capabilities implemented.
- ✓ Full parallel sysplex exploitation, Eliminate Single Points of Failure.
- ✓ Planned outage avoidance (including non-disruptive maintenance implemented.
- ✓ Unplanned outage mitigation.
- ✓ ARM and SFM up to date and automated Restarts, SSDP BCPII, XCF functions implemented.
- ✓ Implementation of resiliency related functions that prevents z/OS /LPAR failures, z/OS Best practices.(SRB and more)
- ✓ Leveraging Sysplex Distributor.
- ✓ Implement Sysplex capabilities and HA for non-IBM products where possible.
- ✓ Follow the CF related best practices (At least 3 physical paths between CF and z/OS LPARs, monitor performance of CF workload, ICF engines..)
- ✓ Be ready for peak workload (LPAR, subsystem level), OOCOD records, TFP_HW, z/OS and subsystem scalability.

❑ Detect/Protect/Monitor

- ✓ Leverage tools to help you increase Resiliency Level
 - ✓ IBM Concert for Z (Including IBM Anomaly Analytics), Middleware AIOPS Products
 - ✓ Syslog analysis (IBM ZLDA ,Observability Connect),
 - ✓ Intellimagic (with build in checks over 600)
- ✓ Strengthen the muscles of Performance Management/Monitoring
- ✓ Improve Capacity Management Processes
- ✓ Leverage **z/OS Health Checker for z/OS and all available product HCs, integrate** z/OS Health Checker exceptions into monitoring process. (Especially GDPS and critical products)

☐ **Business Departments Requirements and Setup**

- ✓ For RTO and RPO, conduct agreements with Business Departments and implement architecture based on requirements.
- ✓ Deploy required solutions to meet business RPO / RTO requirements
- ✓ Factors: government regulations, data center locations, site redundancy, storage resiliency, etc.
- ✓ Deployment trends
 - 2 sites / 2 copies of data => 3 sites / 3 copies of data (optionally extra copy per site for DR testing)
 - 3 sites / 3 copies of data => 4 sites / 4 copies of data (optionally extra copy per site for DR testing)
 - Stand-in capability

☐ **Deploy required solution to meet cyber resiliency requirements**

- ✓ Factors: government regulations, LoB data requirements, etc.
- ✓ Deployment trends: moving beyond initial exploitation (such as data validation and surgical restoration)

☐ **Automation**

- ✓ Leverage automation where possible.
- ✓ Leverage tools. Automate IPLs, procedures
- ✓ Use GDPS automation capabilities. Decrease human interaction

Resiliency Best Practices

❑ Operational Processes

- ✓ Document Critical Procedures, Critical Recovery Procedures, Test them.
- ✓ Prepare Recovery Procedure if each possible situation and test them.
- ✓ Proactive general preventive maintenance strategy for SWs
- ✓ Install quarterly RSU at least 2 times per year, Watch for HIPERs and PEs in the interim, Refer to z/OS Maintenance Strategy Best Practices White Paper. <https://www.ibm.com/support/pages/node/7145227>
- ✓ Subscribe to Security Portal. Apply security fixes.
- ✓ Practice, practice, practice : Local Recovery , DR recovery...
- ✓ Conduct special team: ' Resiliency' SME from client and IBM. Conduct IT Risk management team if it does not exists.
- ✓ Conduct internal SPOF analysis review studies periodically. (At least once in a year).
- ✓ Prepare methods to calculate cost of outages.

❑ Application Infrastructure Dependency

- ✓ Document Application – Infrastructure Relationships(LPARs,middlewares),
- ✓ Document Application High Level Flows which shows those interactions. Share it reachable by all mainframe teams
- ✓ Control and prevent hard coded dependencies to components. Remove any kind of affinities.
- ✓ Leverage tools to discovery dependencies. ADDI , Watson X Code Assistance, CICS Interdependency Analyzer
- ✓ Leverage TCP/IP SMF 119 statics by all mainframe teams (Real time and long term statistics)

❑ Improve Testing Process

- ✓ Stress test applications: (New applications), Significant application upgrades
- ✓ Consider leveraging TST Solution and Test Licensing Options

It is a journey, Start now !

Have an ongoing success plan, education and skill gaps. Conduct projects to target improving the resiliency level of the platform.

- ❑ New Redbook contains several recommendations and considerations about best practices in all chapters.
- ❑ Specifically, Appendix C of New Redbook : Getting started with Business Continuity on IBM Z summaries best practices for Infrastructure and z/OS and Appendix A describes DC considerations

[Getting Started with Business Continuity on IBM Z](#)

Appendix A. Data center resiliency considerations

Appendix B. Sustaining acceptable performance

Appendix C. Best practices: Reducing risk

C.1 Infrastructure layer

C.2 Operating system (z/OS) layer

Draft Document for Review July 30, 2026 11:51 am SG24-8446-01



Getting Started with Business Continuity on IBM Z

Bill White
Diego Bessone
Abi Bettel-Shaffer
Nino de Carvalho
Marc Coq
Kris Dzialek
Mike Gonzales
Kevin McKenzie
David Raften
Karen Smolar
Meral Temel



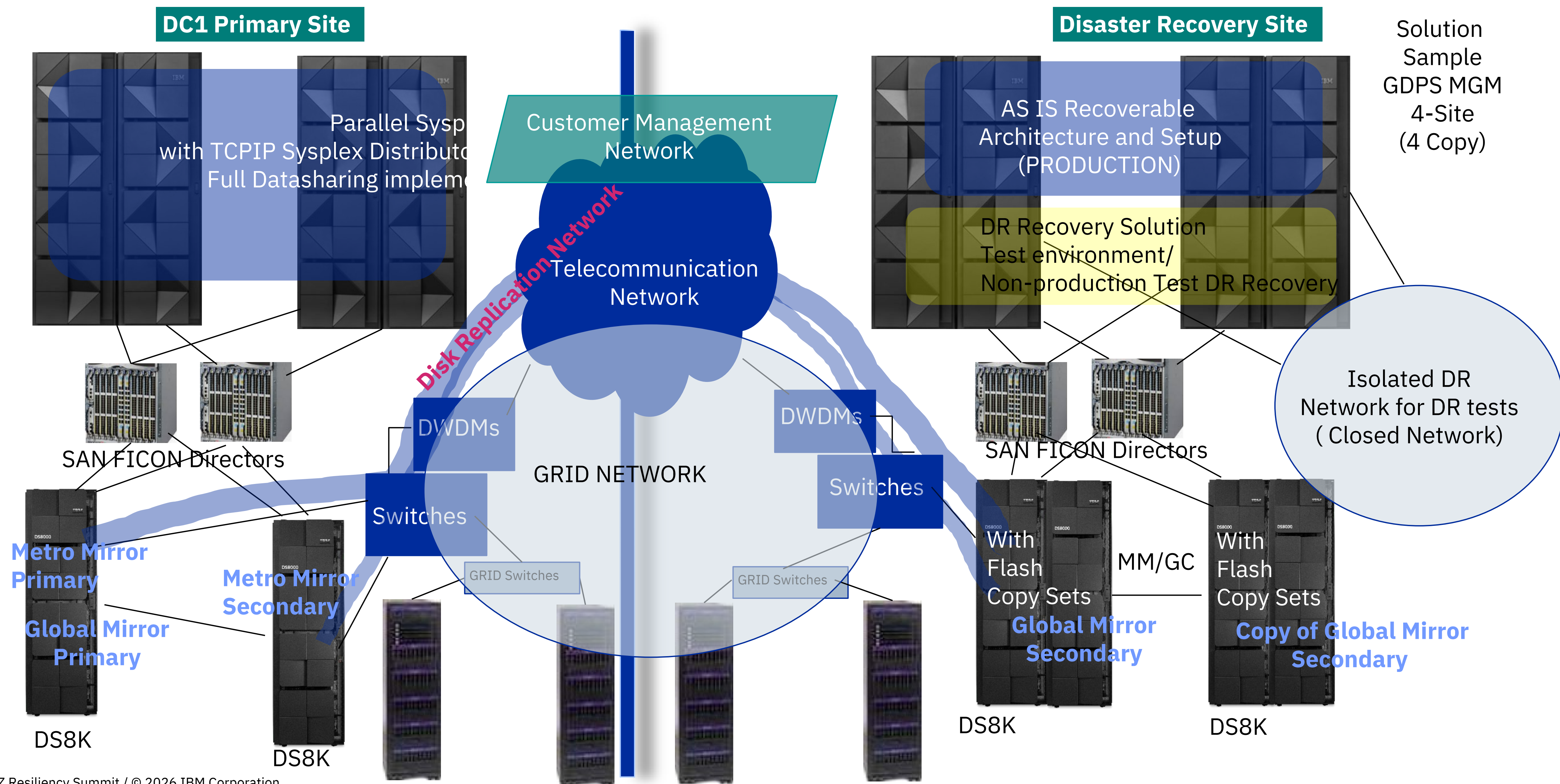
IBM Z

Infrastructure Solutions



BACKUP SLIDES

Infrastructure Best Practices



Common issues : Backup Plans – Be ready for a problem – Test Recovery Processes

Several Leadership Lessons To Learn From This Story...
Try to get ready for Possible Failures, What If Senarios

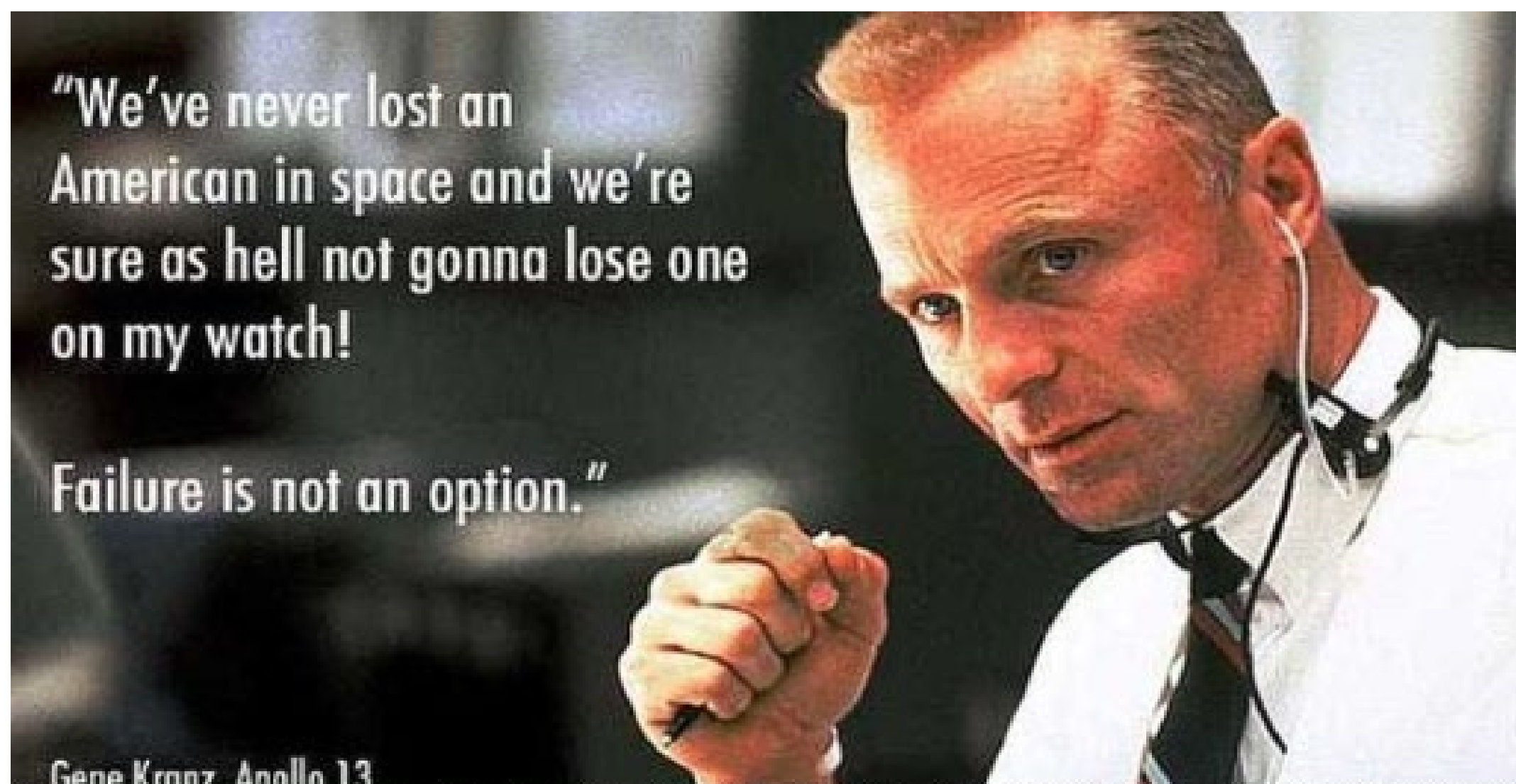
They had plan B,C,D,E.
They had place to test the plans!



Always Have Plan B – Backup Plans!
There are series of methods To Let you a Success Failure!

https://www.youtube.com/watch?v=s_7PfocHTmc

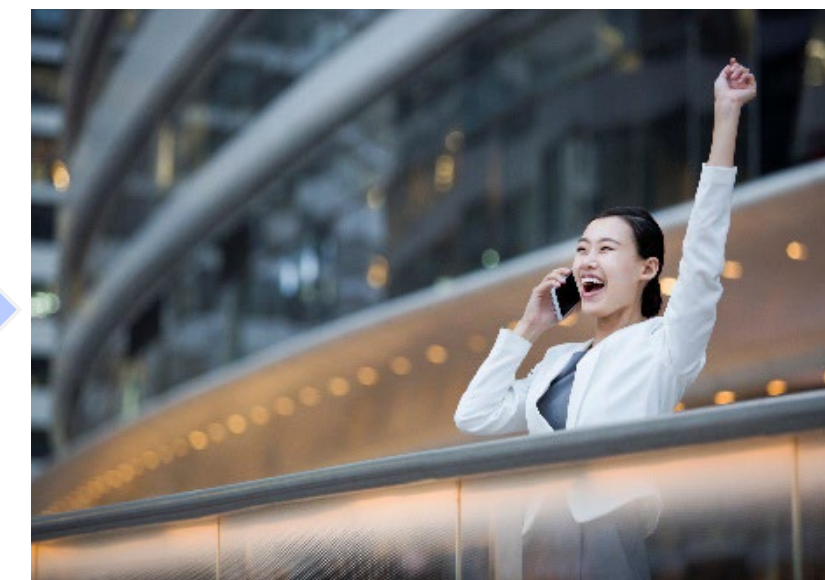
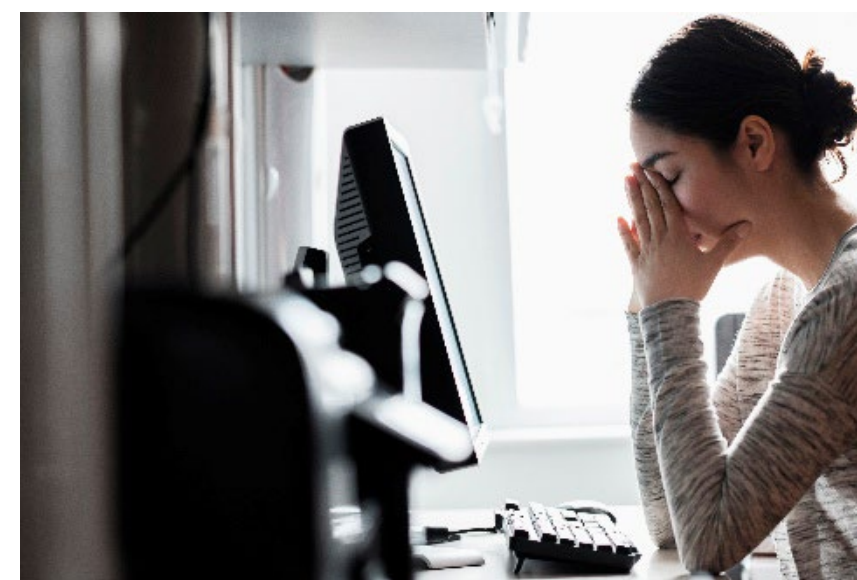
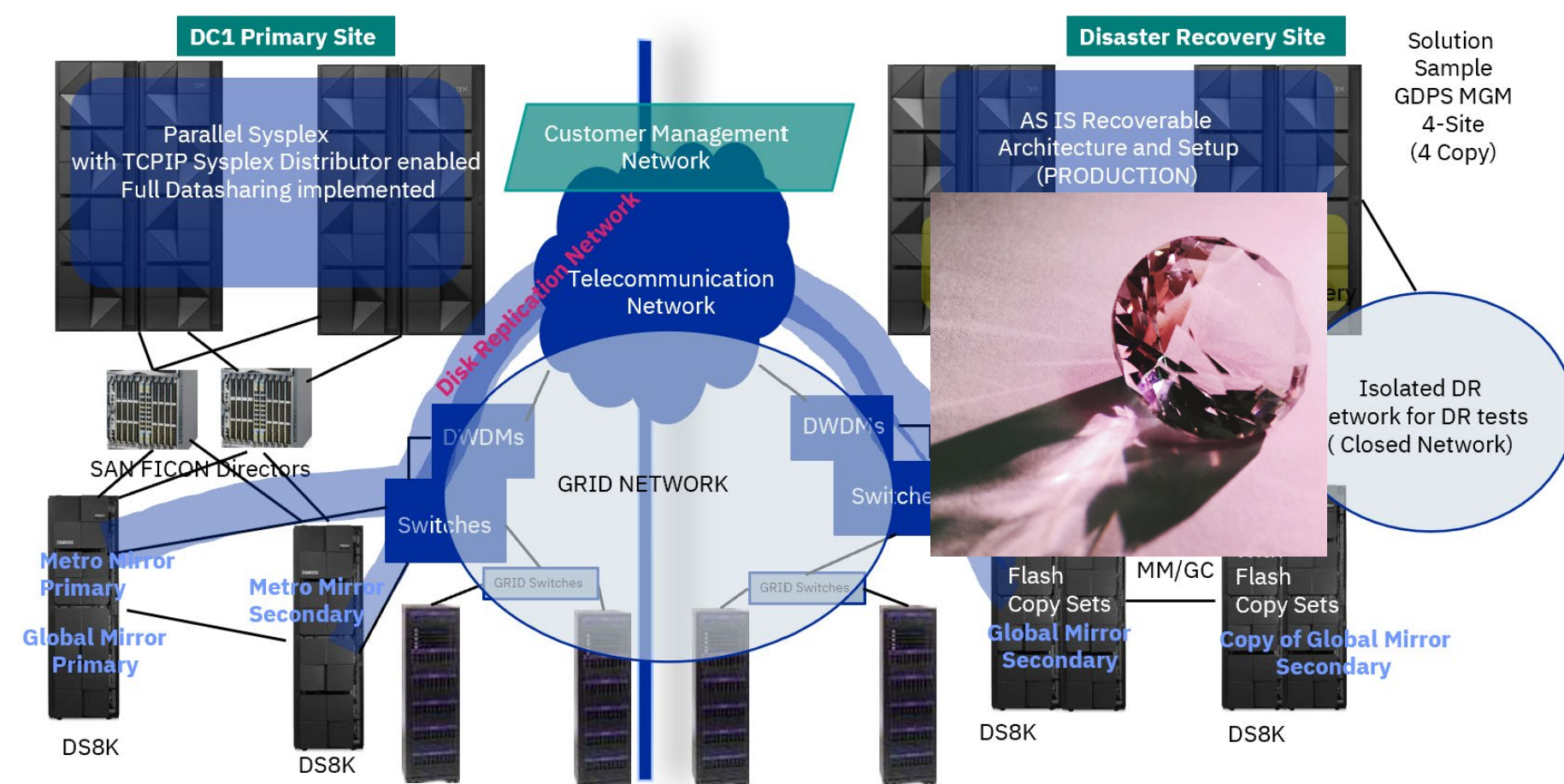
<https://www.forbes.com/sites/geoffloftus/2013/04/03/apollo-13-lessons-from-the-successful-failure/#1fceb25c5d0b>



Common Issues : Disaster Recovery Site gaps

❑ DR site gaps

- Importance and value of Disaster Recovery Site not well understood. (It is not there for just natural disasters or if DC fails)
 - It is not a task that you have to do just to say ‘ Yes, I have DR, yes I am doing tests’. It is very valuable for team and company not only related to ‘natural disasters’ or DC site failures.
 - There are clients who tried to move to DR site not related to natural disaster or Primary DC outages!.
 - Two regions should be smettic from resiliency perspective
 - Should perform DC switch. Example region swap
- When you can not recover on primary site for certain time for any reason, you may go for DR site to continue serving your business. It is not about ‘natural disasters
- DR tests are great opportunity for teams to experience on recovery actions , speeds up troubleshooting on primary site, decreases locally recoverable outages. Flexible capacity solution gives you chance to do more tests and stay longer!.



Importance of Data Center Properties & Capabilities & Tier Certifications

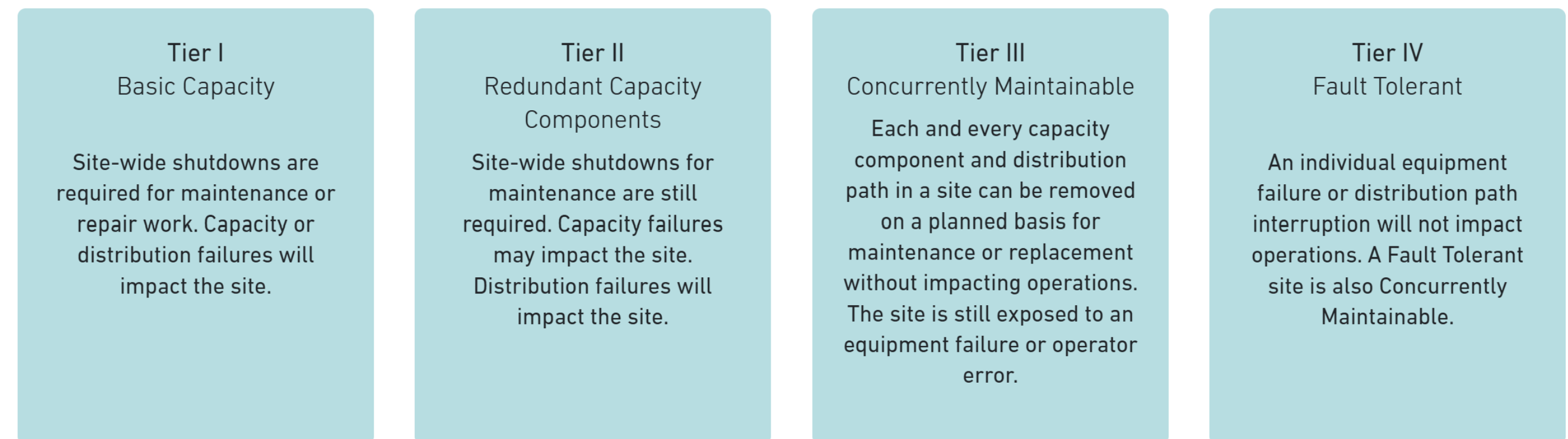
- ❑ Datacenter capabilities impacts the Resilience Level significantly
- ❑ Power outages/ wrong connection/SPOF UPS issues/cooling problems
- ❑ Cabling problems because of mature cabling architecture



❑ Tier level

❑ Operational Tier Level

❑ Same room , same corridor,....

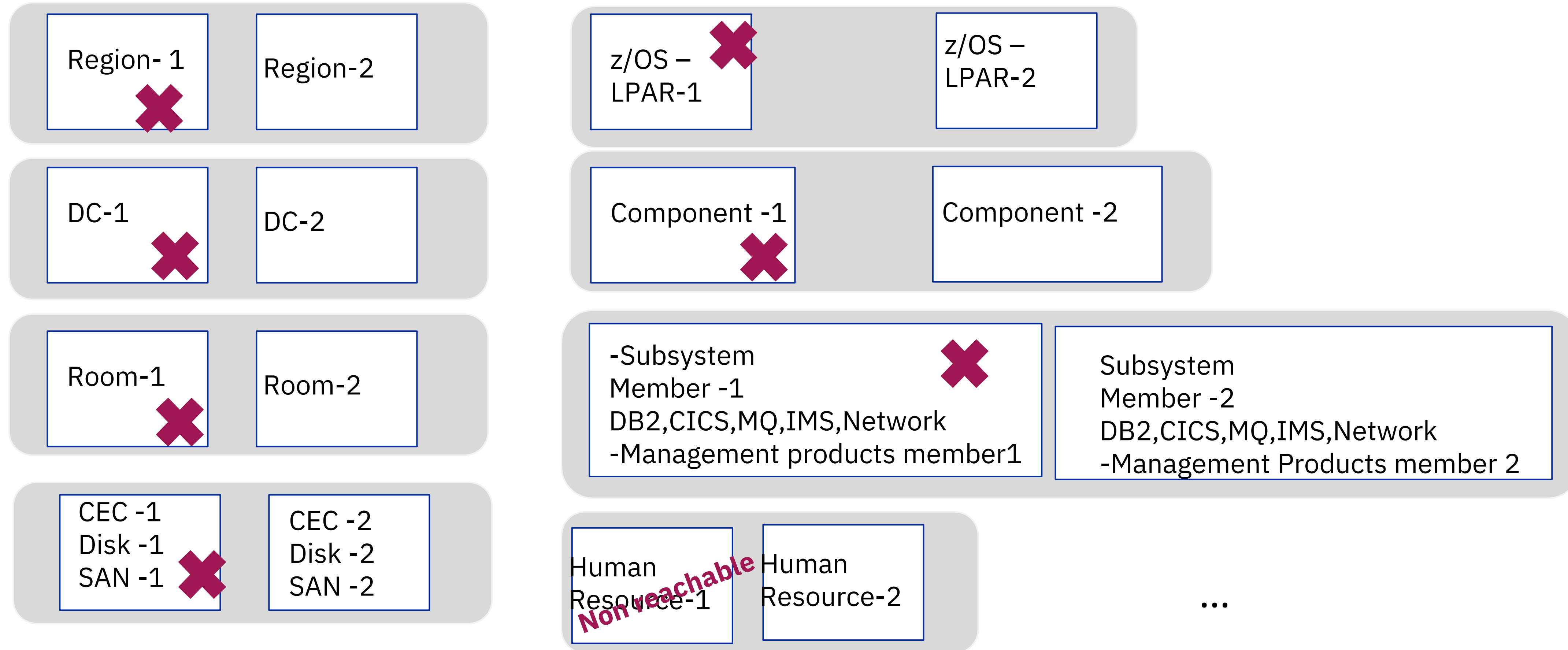


- Datacenter maturity level related outages are very common although they can be prevented
- Actions are done after an outage in several cases.
- Mainframe IT technical teams may not be in authorized persona to decide on big changes but they can definitely raise the risks and may trigger a change

SPOF – Single Point of Failure Analysis

SPOF analysis for HW, SW, human resources, components in all layers....

If DC is impacted, if DC room impacted, if CEC fails, one lpar, one subsystem member fails ...



- Sysplex capabilities fully implemented for every application and products that supports HA.
- SPOF reviews integrated into 'change management' processes, CAB reviews
- Internal team meetings at least once in a year – internal SPOF analysis review studies
- SPOF review in design process, new architectural, infrastructure, application changes...

Common issues: Lack of It mainframe teams – business departments communication

❑ IT mainframe teams – business departments communication

- SLAs are not agreed with business departments .
- Business departments expecting 0 downtime, but IT setup is not aligning with that. In most cases, this is realized after an outage.
- It is important to have SLA agreements with business departments so that investment to resiliency projects can be supported as well. SLA agreements need to be agreed for each business channel.
- Capacity planning process does not include input from business departments growth projections.
Example: Increase in number of ATMs, number of internet banking logins, ...



Common issues: z/OS Health Checker

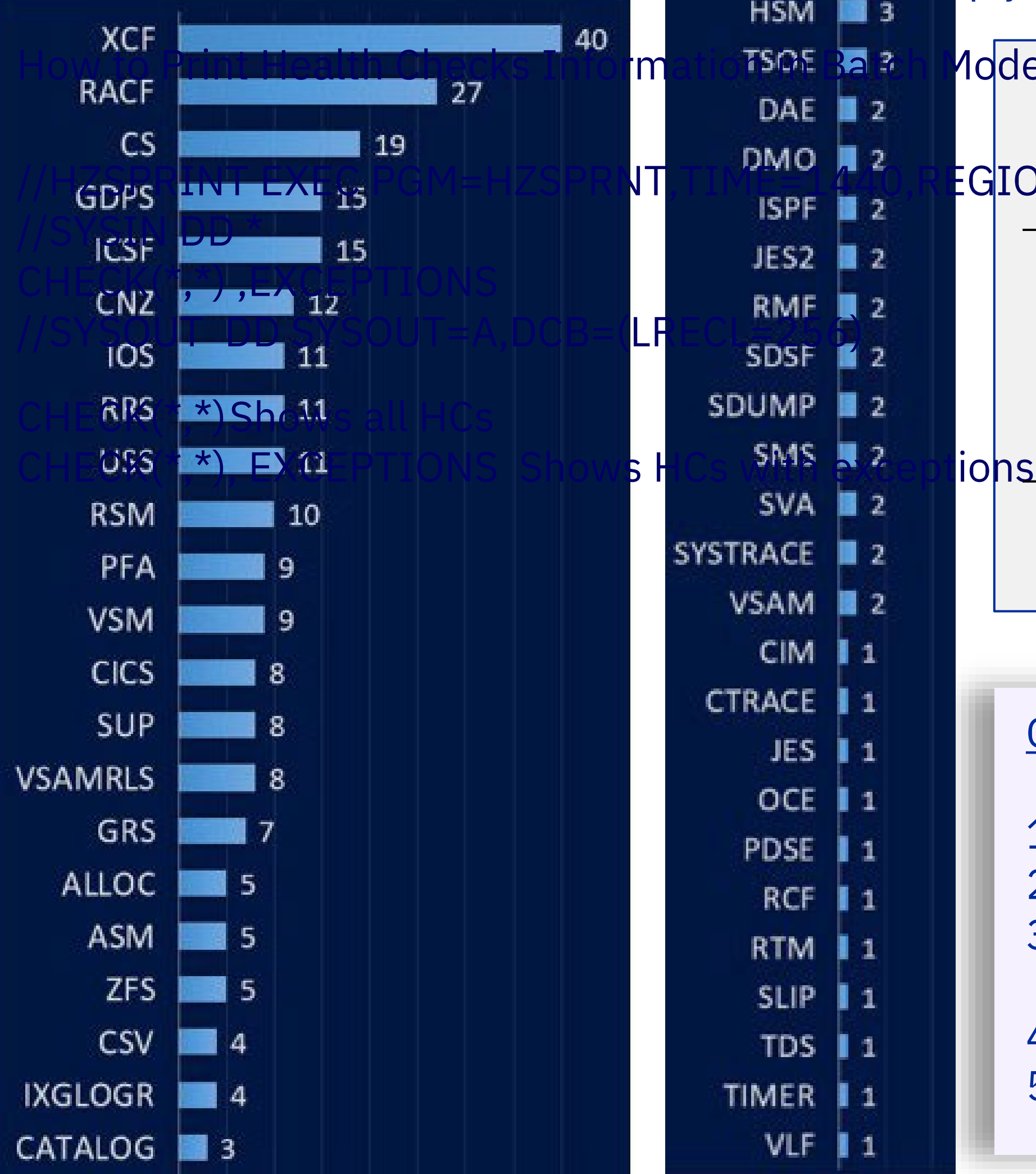
- ❑ Z/OS Health Checker exceptions were not integrated into monitoring processes
- ❑ z/OS Health Checker severity levels lowered from high sev to low sev by clients.
- ❑ Critical z/OS HC exceptions were not recognized and actually they were the ones that warned client to prevent the outage they had experienced.
- ❑ One client had health checks(*,*) inactive in HC policy -> This means they are not using it.



Common issues: z/OS Health Checker

Of Health Checks Group By Owner

Close to 300 Base Health Checks exist that will help you improve your resiliency.



Batch Mode SYS1.SAMPLIB(Hzsprint)

```
//HZSPRINT EXEC PGM=HZSPRINT,TIME=1440,REGION=0M,PARMDD=SYSIN
```

Common Issues : Questions to ask

- 1.Are you tracking IBM.Function.Health Checker fixcat ?
- 2.Are you monitoring HC alerts ? Do you have automation for HC alerts ?
- 3.When did you last check your policy to make sure you activated back HCs temporary deactivated before?
- 4.Did you deactivate or decrease the priority of critical HC – I know let me check that later -?
- 5.What is the default setting for new HCs according to your policy ?